

Ethan Yang

Software Engineer | Pho Enjoyer | Toronto, Ontario | (647) 336-9608 | ethn.yang@mail.utoronto.ca | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Education

University of Toronto

Expected Graduation: May 2029

Bachelor of Applied Science in Computer Engineering

Toronto, Ontario

- **GPA: 3.61**
- Relevant Coursework: Algorithms and Data Structures, Operating Systems, Computer Architecture, Software Engineering, Software Design and Communication, Introduction to Machine Learning

Work Experience

Sun Life Financial

May 2026 – August 2026

API Development Intern

Toronto, Ontario

- Eliminated manual service setup for **8 engineering teams**, cutting a **2-day** process to under **3 hours**, by single-handedly designing and shipping a **Java** platform that automates **Kafka Streams** and **API** infrastructure
- Automated a core piece of the code review process, cutting manual PR review time by an estimated **50%**, by building and deploying an **AI** agent with **AWS Bedrock** that flags documentation violations and drafts fix explanations
- Cut documentation lookup time by over **80%** by building a **Python** and **Node.js** pipeline to migrate **1,200+** legacy pages into structured, searchable **Markdown**

MannLab

January 2026 – Present

Undergraduate Researcher

Toronto, Ontario

- Surfaced unsafe failure modes across **9,144 EEG data samples** by auditing **8 machine learning techniques** and building a **3-step diagnostic tool** in **PyTorch** to trace root causes
- Co-authored **3 research papers** under review at **IEEE ISMAR** and **NeurIPS** by applying signal-processing and **ML** methods to **EEG** biofeedback and consumer BCI safety research under Professor Steve Mann
- Built a real-time **XR** application in **Unity** targeting a **30%** reduction in cocaine cravings by processing streaming **EEG** signal data through a **PyTorch** inference pipeline

Robotics for Space Exploration

September 2025 – April 2026

Software Developer

Toronto, Ontario

- Cut unsafe arm command errors by **40%** by debugging an intermittent real-time communication fault and writing fault-tolerant **ROS2** nodes in **Python** on **Linux**, backed by unit tests
- Scored **93.96%** on the System Acceptance Review at the University Rover Challenge (**20+ points** above cutoff) by building and testing a **PyQt** rover control GUI integrated with **ROS2**, validated against **C++** simulation environments

Projects

Sinatra | [Source Code](#) | [Demo](#)

React | TypeScript | FastAPI | Python

- Attracted a **40+ user waitlist** by independently designing and building a browser-based **DAW** that transcribes vocals to **MIDI** via ML, with AI chord generation and **Three.js** export
- Eliminated install friction entirely by architecting a full in-browser audio pipeline from mic input through multi-track **MIDI** rendering to **FFmpeg** export

Finance Digitization | [Source Code](#)

Next.js | TypeScript | AWS | PostgreSQL

- Cut cheque approval time from **3 days to 1 hour** for **300+ UofT EngSoc users** and **100+ clubs** by building a production system with **OCR**-based receipt parsing and a dual e-signature workflow

Lock Block - HackHive 2026 Winner | [Source Code](#) | [Demo](#)

Python | Flask | OpenCV | Arduino

- Won HackHive 2026 by building a smart lock combining **OpenCV** face recognition with **Ed25519** cryptographic wallet authentication and automated alerting
- Prevented physical bypass attacks by designing a fault-tolerant **Arduino** servo locking mechanism synced to the vision backend via **PySerial**

HATSEYE - DeltaHacks 12 Finalist | [Source Code](#) | [Demo](#)

Python | Flask | OpenCV | Arduino

- Placed **top 6 of 144 teams** by building a dual-pipeline voice-activated AI vision assistant running real-time camera analysis and TTS alongside ultrasonic haptic feedback

Leadership

UofTHacks

March 2026 – Present

Event Logistics Executive

Toronto, Ontario

- Secured a sponsor partnership for UofTHacks 14 by coordinating outreach to external clubs and companies while managing event logistics, guest scheduling, and room allocations

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, C++, C, SQL, HTML5, CSS

Libraries/Frameworks: React, Next.js, Node.js, FastAPI, Flask, PyTorch, OpenCV, Three.js, PyQt, Tailwind CSS, Unity

Developer Tools: Git, GitHub Actions, Bitbucket Pipelines, Jenkins, Docker, Linux/Unix, AWS (ECS, S3, Bedrock), Apache Kafka, Apache Flink, ROS2, Postman, Jira, Arduino, PySerial, Vite